The Paleomagnetism of Diabase Dikes from the Grenville Province

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A paleomagnetic study was attempted of the diabase dike swarm intruding the Grenville structural province of the Canadian Shield. Both the alternating field and thermal demagnetization studies indicated that the dikes have varying degrees of stability of magnetization. Some dikes were shown to have stable primary component of magnetization. Microscopic observations indicated that, in most cases, the primary magnetic mineral titanomagnetite was altered to titanomaghemite, probably due to low temperature oxidation. A good correlation is obtained between magnetic stability and relative abundance of titanomagnetite to titanomaghemite in that dike. The mean paleomagnetic pole for the Grenville dike swarm, which is probably of late Precambrian age, was computed to lie at 3.0 ° N, 29.0 ° W with $A_{95} = 11.0$ °. Comparison of this result with other Precambrian pole positions of similar age relative to North America suggests that the dikes were intruded after the Grenville orogeny.

Une étude de paléomagnétisme a été entreprise sur les dykes de diabase de la province de Grenville. La desaimantation thermique ou par champ alternatif, a montré que les dykes avaient divers dégrés de stabilité d'aimantation. Quelques dykes ont montré une aimantation primaire stable. Les observations microscopiques indiquaient que, dans la plupart des cas, le minéral magnétique primaire, qui était la titanomagnétite, était altéré en titanomaghémite, probablement à la suite d'une oxydation à basse température. Il esiste une bonne corrélation entre la stabilité magnétique et l'abondance relative de titanomagnétite et de titanomaghémite dans ce dyke. Le pôle paléomagnétique moyen pour les dykes de la province de Grenville, qui sont probablement du Précambrien tardif a été déterminé à 3.0 ° N, 29.0 ° W avec $A_{95} = 11.0$ °. Comparé à d'autres positions des pôles précambriens du même âge en Amérique du Nord, ces résultats suggèrent que les dykes se sont mis en place après l'orogenèse Grenville.

Introduction

Diabase dike swarms are numerous across the Canadian Shield and are observed in all structural provinces. These dikes are well distributed in time and space and are very informative regarding the tectonic development of the Canadian Shield (Fahrig and Wanless 1963). Of particular interest are the diabase dikes in the Grenville structural province. The Grenville province lies on the southeastern edge of the Shield and represents a region in which orogenic activity took place one billion years ago. These dikes strike roughly east–west and occur throughout Ontario and Quebec as evidenced from the extensive aeromagnetic maps available of this area. According to Wilson (1965), the time of intrusion of these dikes is related to the mountain building activity of Grenville times. Potassium–argon age determinations by the Geological Survey of Canada have given ages ranging from 450 m.y. (65–113) to 974 m.y. (Wanless *et al.* 1967). Strangway (1964) quotes a radiometric age of 790 m.y. for these dikes. In addition to the potassium–argon age determinations, the only other control on the ages of these dikes is the evidence cited by a few geologists (*e.g.* Gill and L'Espearance 1952; Wilson 1965) who noted that the dikes had cut into 'Grenville strata' and hence concluded that the dikes were of late Precambrian age. The possibility cannot be ruled out that the dikes were intruded earlier than the Grenville orogeny and that the low apparent ages are caused by leakage of argon during later alteration. However, it is unlikely that the dikes are pre-Grenville as they show no signs of metamorphism either in hand samples or polished sections.

If the dikes were indeed younger than the Grenville metamorphism and their true ages could be established, they would be of special interest because the polar wander curve relative to North America is completely unknown in the age range 1000 to 400 m.y. The paleomagnetic study was also undertaken with a view to test Strangway's (1961) hypothesis that the stable

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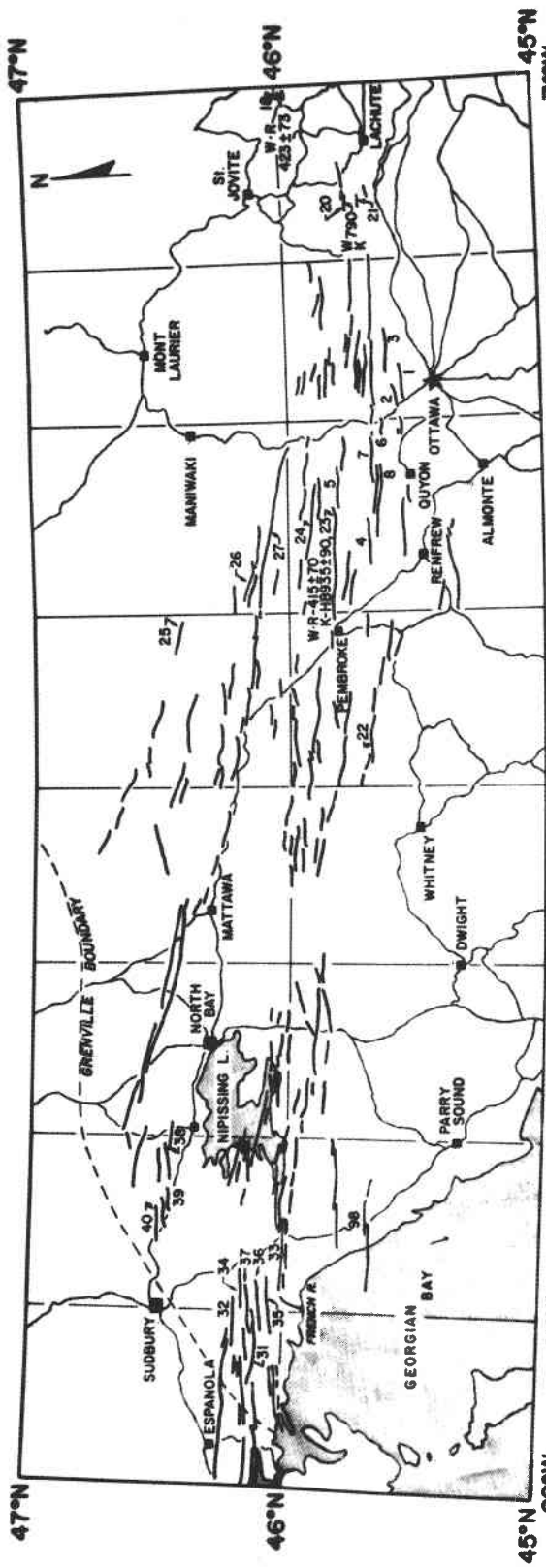


FIG. 1. Location of diabase dikes intruding the Grenville province of Ontario and Quebec.

component of magnetization in dikes tends to lie close to the plane of the dike. Evidence against this hypothesis was later advanced by Fahrig *et al.* (1965), by Evans (1968) and more recently by Larson and Strangway (1969). Both as a precaution in paleomagnetic study of the dikes and for the inherent interest of the point, it would be a useful corollary to test the data for consistency with Strangway's original hypothesis.

The dikes intrude throughout the Grenville geological province of Ontario and Quebec. The composition of the rocks is of the quartz diabase type. These dikes are often several hundred kilometers long (as inferred from aeromagnetic maps) and are usually 20–150 m wide. Initial sampling of nine dikes was made by Dr. W. F. Fahrig of the Geological Survey of Canada who kindly made the samples available to the author. In view of the vast area in which these dikes are exposed, samples from nine sites cannot adequately represent this dike swarm. Accordingly, further sampling was done by the author during the summer of 1968. In all, one hundred and thirty-seven samples were collected from twenty-eight dikes whose locations are shown in Fig. 1. Sites 1–28 are from the eastern part of the Grenville Province while sites 31–37 are from the western edge of the Grenville front. Sites 38–39 and 98 are from the region in between. The cores were oriented *in situ* by means of a solar compass similar to that described by Gough and Opdyke (1963). Each dike is taken to represent one site for statistical analyses in this study. At least five samples (cores) were collected from each dike and wherever possible these samples were equally spaced across the width of the dike. The samples were in the form of cylindrical cores 2.58 cm in diameter and about 15 cm in length. Two cylinders 2.58 cm long were cut from each core and used in the paleomagnetic study.

Remanent Magnetization

The direction and intensity of remanent magnetization of each specimen was measured by means of a PAR spinner magnetometer. The vector sum of the two unit vectors representing the magnetization of two cylinders from a given core was calculated and taken as the mean direction of magnetization for that core. If the vectors of magnetization diverge by an angle exceeding 23°, they were termed 'inhomogeneous' and were not included in any further analysis. This is a procedure originally suggested by Larochelle (1966) and is based on a minimum within core precision of 25. The magnetization of 72% of the cores is 'homogeneous' before demagnetization. The mean direction of magnetization for the dike was taken as the vector sum of N unit vectors representing the N cores from that dike. The NRM moments average to 1.74×10^{-2} G cm^3 and were distributed between 4.5×10^{0} and 5.4×10^{-6} G cm^3. The results are given in Table 1 and plotted in Fig. 2. The grouping in Fig. 2 suggests the presence of a strong secondary component masking any primary component of magnetization of the dikes.

Systematic demagnetization by alternating fields was carried out in an attempt to isolate the primary component of magnetization. Demagnetization was conducted on two pilot specimens from each site by means of an apparatus similar to that described by McElhinny (1966). Not all sites yielded stable end-points of direction during demagnetization; hence a compromise procedure between that of McElhinny and Gough (1963) and of Irving *et al.* (1961) was adopted to decide on an optimum field of demagnetization. If a stable end-point in direction was obtained

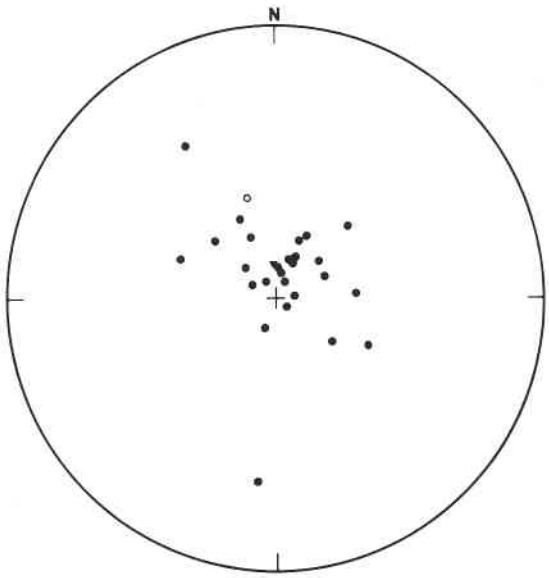


FIG. 2. Mean site directions of Grenville dikes: natural remanent magnetization.

TABLE 1. Paleomagnetic data of Grenville dike swarm

Dike	M_N				B_{ac}	M_{ac}					Site pole	
	N	D	I	K	Oe.	D	I	R	K	α_{95}	Lat.	Long.
GR01	5	(335½)	(55)	2		random						
GR02	5	67	67½	4	100	103	61½	4.803	20	14	21½ N	25½ W
GR03	5	297½	79½	7	50	2½	85½	4.792	20	14	55 N	75 W
GR04	6	(343½)	(−48½)	2	150	296	−61	5.771	22	12	15 S	146½ E
GR05	5	51	66	3	100	349½	78	4.879	33	11	68½ N	88 W
GR06	5	(26½)	(61)	2		random						
GR07	6	338½	63	18	100	327	76½	5.738	19	13	64 N	107½ W
GR08	5	(25½)	(72)	2	200	107	−44½	4.713	14	17	30 S	22 E
GR18	6	200½	76	16	150	199	82	5.809	26	11	31½ N	80 W
GR20	5	17	80	13	150	183	65	4.748	16	16	3 N	76½ W
GR21	3	127½	60½	11	150	139½	14	2.918	24	16	26 S	29 W
GR22	6	129½	83½	5	300	167½	−57½	5.113	6	24	78 S	20 W
GR23	5	31½	82½	26	100	123½	83½	4.704	14	17	37½ N	63 W
GR24	3	291½	48	15		random						
GR25	4	(117)	(48½)	3	250	118½	58	3.937	47	10	11½ N	33 W
GR26	4	27½	72	34	100	20½	57	3.746	12	20	66 N	13½ W
GR27	3	(32)	(10½)	2	300	337	66½	2.883	17	19	74 N	147½ W
GR28	4	(82½)	(82)	2	150	104	−63½	2.698	7	31	39 S	42½ E
GR31	5	(46)	(49½)	3	200	114½	44½	4.001	4	31	3½ N	26 W
GR32	5	29	74	17	250	117½	27½	4.924	53	9	7½ S	21 W
GR33	5	87	57	20	100	97	50	4.591	10	20	17 N	18 W
GR34	5	(329)	(24)	2		random						
GR35	6	(186)	(22)	1	400	114	33	4.417	3	32	2½ S	21 W
GR36	5	24	64	26	200	(106½)	29	4.987	299	4	½ N	14 W
GR37	5	314½	72	23	300	119	36	4.971	140	5	4 S	25½ W
GR38	5	5	77½	16		random						
GR39	5	312½	56	8	100	320	68½	4.024	4	30	63½ N	145 W
GR98	6	331½	83	14	100	284½	66½	5.421	9	20	41 N	138 W

NOTE: N = number of samples measured; R = length of the resultant of the N unit vectors; D = declination of remanent magnetization; I = inclination of remanent magnetization; M_N = NRM; M_{ac} = RM after demagnetization in alternating field B_{ac}; $K = (N\text{-}1)/(N\text{-}R)$ = estimate of Fisher's κ parameter; α_{95} = semi angle of cone of 95% confidence for mean direction.

during demagnetization, the field at which this was achieved was taken as the optimum field for that site. However, if no stable end-point was observed, which was the case for most of the sites, the specimens were all demagnetized in steps, the precision parameter κ (Fisher 1953) was calculated for each successive step and the field at which the maximum κ was obtained was chosen as the optimum field of demagnetization for that site. The optimum field for most of the sites was about 100 Oe indicating a magnetic mineral of low coercivity. Ten of the twenty-eight sites have random NRM directions on Watson's (1956) criterion. The NRM directions of these sites are shown in parenthesis in Table 1. The grouping of site GR01 improved to being barely significant after demagnetization while that of sites GR06, 24, 34, and 38 was not improved. The individual specimen vectors for the latter sites moved about in a random fashion; large fluctuations in intensities were also noticed. Such behavior is typical of unstable remanent magnetization and results from these sites were therefore not included in the final analysis. In the rest of the sites, demagnetization improved the grouping as may be noticed from increases in the values of κ. In general, specimens of the Grenville dikes behaved in three different ways during demagnetization which formed the basis for dividing them into the three following groups:

Group 1

For some of the sites which had NRM directions close to the present earth's field, demagnetization improved the grouping but with insignificant changes in the mean direction. No stable end-point in direction was observed during demagnetization in successively higher fields. This behavior indicates either that the primary component of magnetization of these dikes has a

direction close to that of the present geomagnetic field or that these dikes do not carry a primary component. Specimens from sites GR02, 03, 05, 06, 07, 18, 20, 23, 25, 26, 27, 39, and 98 behaved in this fashion. Typical directional and intensity changes in magnetization for this group of sites are presented in Figs. 3*a* and 3*b*.

Group 2

All specimens from sites GR04, 08, 22, and 28 showed a systematic change in direction during demagnetization. Sites which originally gave directions with high positive inclinations moved systematically in such a way that stable end-points in direction were obtained with steep negative inclinations. Better within-site grouping was observed after demagnetization. Changes in direction and intensity typical of this group are shown in Figs. 3*c* and 3*d*.

Group 3

A more conventional behavior was observed for specimens from the remaining sites. On demagnetization, specimen magnetization directions were found to move to easterly azimuths and shallower inclinations. Such behavior is to be expected if a viscous component due to the present geomagnetic field was being eliminated during demagnetization, and a stable component was retained. Typical examples of this directional behavior are shown in Fig. 3*e* and the corresponding normalized intensities in Fig. 3*f*. It may be noted that there is no marked difference between the intensity curves for the three groups selected on their directional changes, except the Group 3 specimens retain a larger fraction of their initial moment to 500 Oe.

In view of the different responses of specimens from the three groups of sites to alternating field demagnetization, it was decided to study the behavior of specimens during thermal demagnetization. At least one specimen from each site, and in some cases three, were selected for thermal treatment. The specimens were heated in air to successively higher temperatures of 150, 200 · · · 610° C and cooled in field free space. It was found that specimens behaved in three distinct ways, and that the sites grouped together on the basis of response to alternating field demagnetization also gave similar responses when heated. It is thus reasonable to discuss the thermal studies under the same groups as before.

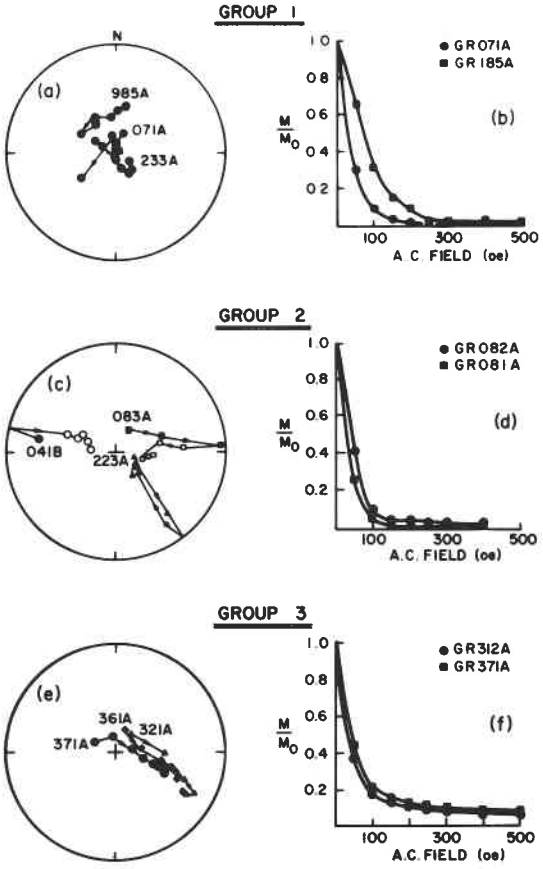


FIG. 3. Response of specimens of Grenville diabase to a.f. demagnetization.

Group 1

The response of four specimens from site GR18 was studied. There is a stable end-point in direction during thermal demagnetization. A similar response was noticed of specimens from all Group 1 sites. The stable directions were maintained up to 550° C after which large movements in direction were noticed, with abrupt changes in intensity. The mean directions obtained for these sites after thermal demagnetization are not significantly different from those obtained after alternating field demagnetization. The changes in direction and in intensity during thermal demagnetization are plotted in Figs. 4*a* and 4*b*, respectively.

Group 2

The response of specimens from sites GR08 and 28 is very similar to that under a.f. demagnetization. A similar systematic change in direc-

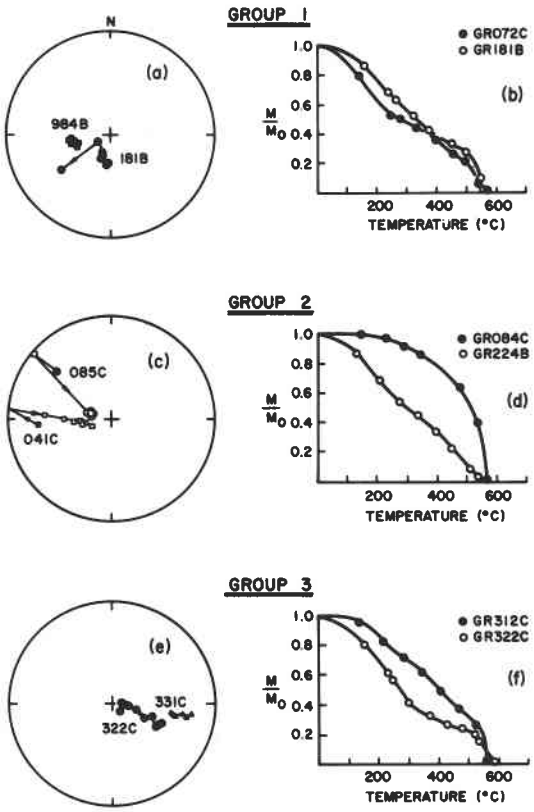


FIG. 4. Response of specimens of Grenville diabase to thermal demagnetization.

tion was observed from positive to negative inclinations. The change in magnetization for this group of sites is shown in Figs. 4*c* and 4*d*.

Group 3

The response of specimens from site GR32 is typical of those from this group. An eastward movement in direction was noticed, the movement being very similar to that caused by alternating field demagnetization. The directional changes in magnetization after each heating are shown in Fig. 4*e* and curves of normalized intensity plotted versus temperature are given in Fig. 4*f*.

Thermal demagnetization studies confirm the conclusions drawn from the observations of alternating field demagnetization. It is very likely that the specimens from the three groups of sites carry primary components of magnetization. However, the specimens from different groups of sites exhibited varying degrees of stability and behaved differently under both alternating field and thermal demagnetization. Such a marked contrast in magnetic properties is expected to be related to the state of alteration of the magnetic mineral.

Opaque Petrologic Properties

Polished sections were made of one sample of known remanence from each of six dikes to study the relations of the iron–titanium phases. All polished sections were observed under an ore microscope at magnification × 160. Systematic measurements were made of VHN number, reflectivity and grain-sizes of the opaque minerals. The former two parameters clearly distinguish between the various magnetic minerals according to the values tabulated by Bowie (1967). Table 2 lists some representative values for these parameters obtained for the opaque minerals in these

TABLE 2. Summary of observations on polished sections of Grenville diabase dikes

Grain size (μ)	VHN	Refl.	Lamellas	Description
		Sample GR181		
75 × 118	562	18	Abundant	Isotropic grain with lamellas
64 × 48	602	20		
48 × 173	611	20	Present	A and B minerals
41 × 39	1022	25		coexisting in same
152 × 49	564	24		grain
71 × 156	551	23	Present	
		Sample GR231		
64 × 37	495	19	Present	A and B minerals
51 × 29	947	23		coexisting in same grain
77 × 46	603	22	Abundant	
23 × 36	572	21	Present	Isotropic
19 × 28	1012	26		Anisotropic
46 × 41	985	27		Anisotropic
		Sample GR221		
95 × 105	737	26		Titanomaghemite
75 × 58	579	22	Present	Titanomagnetite
52 × 67	848	24		Anisotropic
48 × 84	651	21	Present	Isotropic
58 × 76	1021	26		Anisotropic
		Sample GR081		
85 × 112	865	23		Extensive lamellas are present in more than 25% of grains
76 × 165	738	24	Abundant	
78 × 92	561	21	Present	Isotropic
62 × 70	982	25		Anisotropic
90 × 76	662	20	Present	Titanomagnetite
46 × 80	985	24		Anisotropic
98 × 48	542	21	Few	

Table 2. Concluded

Grain size (μ)	VHN	Refl.	Lamellas	Description
		Sample GR364		
19 × 37	572	19	None	A and B minerals
7 × 21	953	23		coexisting in same grain
20 × 42	602	21		
26 × 46	485	21	Few	
44 × 29	512	21	Few	
		Sample GR374		
135 × 106	555	22	Few	Slightly anistropic
47 × 52	607	19	None	Isotropic
55 × 38	605	21	None	A and B minerals
16 × 24	1055	26		coexisting in same grain
19 × 37	572	19	None	A and B minerals
7 × 21	953	23		coexisting in same grain

sections. Two types of minerals (named A nd Ba in Table 2) are distinguishable. The mineral A is identified as titanomagnetite from its gray color, isotropic features, typical values of VHN, and reflectivities. The mineral B, which is slightly bluish-gray in color, often anisotropic with higher values of VHN and reflectivity was identified as titanomaghemite. Both minerals were found to coexist in all sections with varying abundance. Titanomaghmite is probably formed by the low temperature oxidation of titanomagnetite. Typical examples of the types of minerals observed are shown in Fig. 5.

As many petrologic parameters as possible were estimated for the six samples in order to attempt a correlation between petrologic parameters and magnetic stability. These include the grain-size, the relative abundance of titanomagnetite to titanomaghemite, and Ade-Hall *et al.*'s (1968) magnetite oxidation number M which represents the state of oxidation of the magnetic mineral. An average value of grain-size and the relative abundance of titanomaghemite to titanomagnetite (called B/A in Table 3) were estimated from the observations for each sample and are listed in Table 3. As each sample of diabase contained grains representing several classes of oxidation, the procedure in calculating M in-

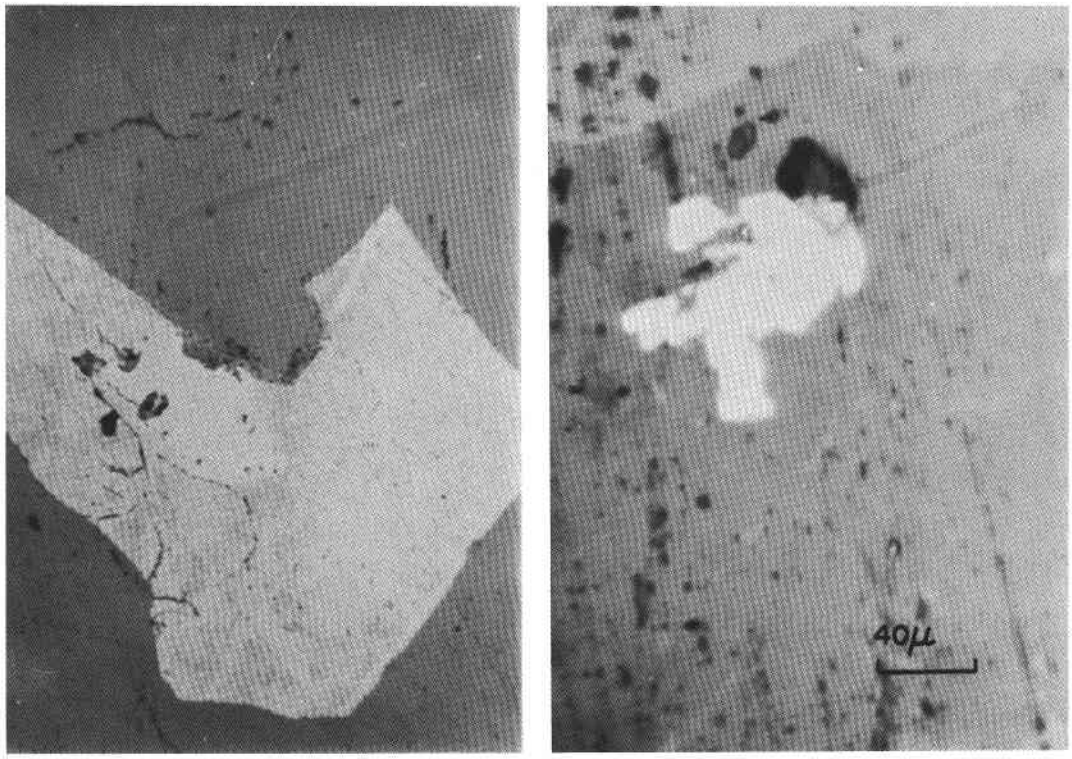


FIG. 5. Photomicrographs of polished sections of Grenville diabase showing (*a*) alteration of titanomagnetite to titanomaghemite in GR181, (*b*) a titanomagnetite grain in GR364.

TABLE 3. Magnetic and petrologic parameters of Grenville diabase

Sample No.	Grain size (μ)	B/A	M	K	SF	S_{200}
GR081	90–100	0.459	2.87	13.9	0.187	0.453
GR181	80–90	0.372	2.38	26.2	0.816	0.899
GR221	60–70	0.520	2.44	5.6	0.261	0.425
GR231	70–80	0.465	3.11	13.5	0.119	0.914
GR364	20–30	0.108	1.28	298.6	7.430	0.613
GR374	40–50	0.197	1.73	140.0	1.360	0.639

NOTE: B/A = relative abundance of titanomaghemite to titanomagnetite; M = Ade-Hall *et al.*'s magnetite oxidation number; K = estimate of Fisher's κ parameter; SF = stability factor defined in text; S_{200} = stability factor of Wilson *et al.* 1968.

volved the following: classification of each part of the grain into one of the six classes of oxidation, measurement of percentage of total magnetite area represented by that class, and fitting these values in Ade-Hall *et al.*'s definition. An average number of more than one hundred opaque grains from each sample, well spread over the polished section, was taken as a representative sample for estimation of the magnetite oxidation number for that sample. These values are listed in Table 3.

There are numerous examples in the literature of the correlation of magnetic properties of the rock with the opaque petrological parameters. Of particular interest to paleomagnetism is the correlation of magnetic stability with the petrologic parameters. Extensive reviews of the existing literature were made by Ade-Hall *et al.* (1968), Larson *et al.* (1969) and others. Different groups of workers have used different parameters as representing magnetic stability. An ideal magnetic stability factor should be representative of the changes in direction as well as intensity of magnetization of the sample. This is particularly important in correlations between petrologic and magnetic parameters. Ade-Hall *et al.* (1968), following an earlier suggestion of Wilson *et al.* (1968), used a stability factor S_{200}. They defined S_{200} as $R_{200}/(R_{200} + r)$ where R is the length of the remanent magnetization vector after demagnetization in 200 Oe and r is the vector change between 0 and 200 Oe treatment. Thus a rock with stable magnetization, according to this definition, will have S_{200} approaching 1.0 while rocks with unstable magnetization will often have S_{200} less than 0.5. This definition of stability factor has the disadvantage in that it is effected by the divergence of the direction of magnetization of the secondary component from the primary component and by the nature of the secondary magnetization. Larson and Strangway's (1969) tabulation of values of κ implies their choice of the Fisher precision parameter as a stability factor. κ reflects the tightness of grouping of the directions of magnetization alone, and takes no account of magnetization magnitudes. Tarling and Symons (1967) introduced a stability factor SI which also has the limitation that it is related only to remanence directions. An alternative stability factor SF which is defined as

$$SF = \left(\frac{M_{opt}}{M_o}\right)/\Theta^{\circ}_{63}$$

was used in this study where M_{opt} and M_o are the values of remanence intensity at optimum field of demagnetization and at zero field (NRM) and Θ_{63} is the circular standard deviation of the directions taken by the magnetization between zero field and the optimum field. Θ_{63} was chosen as representative of the directional change of remanence because it is the least sensitive to the number of included directions of the most commonly used measures of spherical distribution (Tarling and Symons 1967). The factor SF may reflect better the stability of magnetization as the directional and intensity changes are both incorporated in this artificial number. The various magnetic stability factors listed above are estimated for Grenville diabase and are listed in Table 3.

The magnetic stability factors are plotted against the various petrologic parameters in Fig. 6. The factor S_{200} did not change significantly with any petrologic parameter and hence was no longer considered. Although, in general, samples with smaller grain-size had higher values for all stability factors, Fig. 6 indicates that high magnetic stability is not entirely due to smaller grain sizes. A good correlation was obtained between B/A and magnetic stability. It seems that the observed scatter of magnetization in certain sites is associated with the presence of titanomaghemite. Finally, a systematic variation of all magnetic parameters with the magnetite oxidation number was noticed. It may be noted here that Larson and Strangway (1969) concluded from their observations that magnetic stability is correlated well with the presence of small grains observed

microscopically. The petrologic studies of Grenville diabase indicated that instability of magnetization of this rock unit is associated with maghemitization. Since there is a tendency for larger grains to show greater maghemitization, it is difficult to associate the stability with either of these properties alone. There is also a correlation between magnetic stability and magnetite oxidation number. The relative contribution of these factors in reducing the bulk coercivity of the sample is not known at present. Moreover, most of the stability factors discussed above are statistical functions of the variation of intensity and direction of magnetization of the sample and do not represent a physical property of the mineral. Thus the above correlations do not demonstrate a physical mechanism but only represent an empirical relation between the magnetic and petrologic observations.

Discussion

From the paleomagnetic and petrologic observations of the Grenville diabase, it appears that the magnetization of some of these dikes has been effected by alteration of the primary magnetic mineral to titanomaghemite. The alteration probably took place during Recent times, so that the primary and secondary magnetizations were acquired at different times. In order to obtain a pole position representative of the time of intrusion of Grenville dikes, certain dikes have to be excluded whose magnetization is likely to have been effected by the alteration. The site mean directions after alternating field demagnetization are plotted in Fig. 7 and listed in Table 1. From the data listed in Table 1, it may be said there is no correlation between the orientation of the dikes and their directions of magnetization. These directions are tested for the possible presence of more than one concentration of directions. The procedure was after Watson (1956), the numerical applications of which are given in McElhinny (1963). The test showed that while Group 3 was significantly different from the other two, the data of Groups 1 and 2 gave no reason to assert that the two vectors were different. Further, there was no geological evidence to believe that these dikes belong to different swarms. There was considerable overlap in directions for the three groups of sites. The reversed directions of sites GR08 and GR28 were not significantly different from the Group 1 sites.

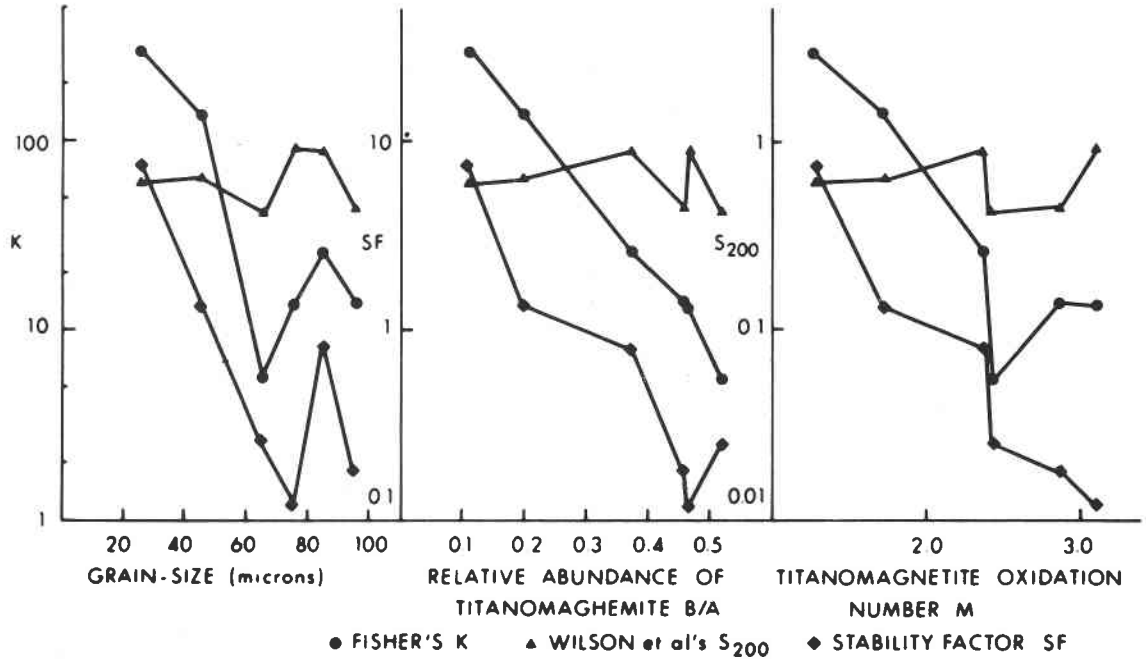


FIG. 6. Plots showing variation of magnetic stability with petrologic parameters: (*a*) Grain size versus magnetic stability, (*b*) Relative abundance of titanomaghemite versus magnetic stability, (*c*) Titanomagnetite oxidation number versus magnetic stability.

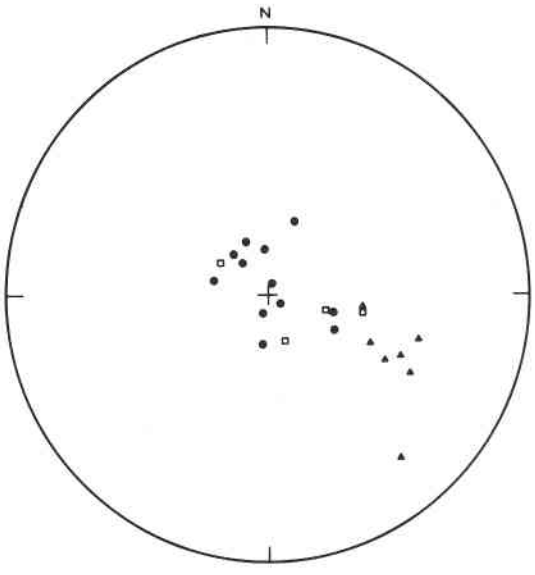


FIG. 7. Mean site directions of Grenville diabase dikes: after optimum demagnetization.

Directions for sites GR02, 04, 20, and 25 were not significantly different from Group 3 sites. Thus, the previous grouping, based on response of specimens to alternating field demagnetization, yields no clear basis for selection of sites. It is possible that the response to a.f. demagnetization depends on the oxidation state, and this in turn on the position of the dike in the Grenville Province. A suggestive point is that all Group 3 sites were collected from the western edge of the Grenville Province. An alternative grouping, based on the remanent magnetization directions, is suggested as follows:

Group A: sites GR02, 04, 20, 21, 25, 31, 32, 33, 35, 36, 37

Group B: sites GR03, 05, 07, 08, 18, 22, 23, 26, 27, 28, 39, 98

This grouping is supported by petrological studies reported in earlier sections. In Group B sites, high B/A values suggest that the primary magnetization has been partly destroyed by recent oxidation of the magnetic mineral to titanomaghemite. Mean directions and pole positions for the two groups of sites are listed in Table 4. These two directions are significantly different at the 95 % level according to an F-test (McElhinny 1963).

Of the two groups of poles discussed above, the mean Group A pole is believed to be representative of the geomagnetic field at the time of intrusion of the dikes. While the age of the Grenville dikes is uncertain, this pole position is consistent with other paleomagnetic data from North American rock units which may be of similar age. However, the comparison is limited by the fact that there are very few paleomagnetic results of comparable age. DuBois (1962), in his chronologic correlation of Keweenawan rock units, reported the paleomagnetic results from granite gneisses in the Township of Digby and from syenitic gneisses in Wakefield County, both in the Grenville Province. The pole position he obtained was at 7.0° S and 162.0° E which is essentially the antipole of the Group A sites of this study. Most of the Keweenawan pole positions reported recently in literature are of rock units older than the Grenville dikes. Hood (quoted in Cox and Doell 1960) has obtained paleomagnetic results for certain intrusive rocks from Bancroft, Ontario, in the Grenville province. The paleomagnetic pole at 8.5° N, 22° W is very similar to the pole for Group A sites. Fahrig *et al.* (1971) have recently obtained a paleomagnetic pole at 8° N, 167° E for Franklin diabase dated at 675 m.y. The closeness of this pole position to the pole for the Grenville dikes, in conjunction with the K–Ar age determinations, suggests that the Grenville dikes were intruded during post Grenville times and most probably around 700 m.y. Any further conclusions of paleomagnetic results should wait until more data are available on the ages of these dikes.

TABLE 4. Group mean directions of Grenville diabase

	N	D	I	κ	α_{95}	Lat.	Long.	A_{95}
Group A	11	118.9	45.2	14.0	11.3	3.0 N	29.0 W	11.0
Group B	12	315.6	75.3	20.7	8.9	61.6 N	110.1 W	14.5

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ADE-HALL, J. M., KHAN, M. A., DAGLEY, P., and WILSON, R. L. 1968. A detailed opaque petrological and magnetic investigation of a single Tertiary lava flow from Skye, Scotland. Geophys. J. Royal Astr. Soc. **16**, pp. 375–415.

BOWIE, S. H. U. 1967. Microscopy: Reflected light. *In* Physical Methods in Determinative Mineralogy. *Edited by* J. Zussman, Academic Press, New York. 514 p.

COX, A. and DOELL, R. R. 1960. Review of paleomagnetism. Geol. Soc. Am. Bull. **71**, pp. 645–768.

DUBOIS, P. M. 1962. Palaeomagnetism and correlation of Keweenawan rocks. Geol. Surv. Can. Bull. **71**, 75 p.

EVANS, M. E. 1968. Magnetization of dikes: a study of the paleomagnetism of the Widgiemooltha dike suite, Western Australia. J. Geophys. Res. **73**, pp. 3261–3270.

FAHRIG, W. F., GAUCHER, E. H., and LAROCHELLE, A. 1965. Palaeomagnetism of diabase dykes of the Canadian Shield. Can. J. Earth Sci. **2**, pp. 278–298.

FAHRIG, W. F., IRVING, E., and JACKSON, G. D. 1971. Paleomagnetism of the Franklin diabases. Can. J. Earth Sci. **8**, pp. 455–467.

FAHRIG, W. F. and WANLESS, R. K. 1963. Age and significance of diabase dyke swarms of the Canadian Shield. Nature, **200**, p. 934–937.

FISHER, R. A. 1953. Dispersion on a sphere. Proc. Royal Soc. London, Ser. A, **217**, pp. 295–305.

GILL, J. E. and L'ESPEARANCE, R. 1952. Diabase dykes in the Canadian Shield. Trans. Royal Soc. Canada, **XLVI**, pp. 25–36.

GOUGH, D. I. and OPDYKE, N. D. 1963. The palaeomagnetism of the Lupata alkaline volcanics. Geophys. J. Royal Astr. Soc. **7**, pp. 457–468.

IRVING, E., STOTT, P. M., and WARD, M. A. 1961. Demagnetization of igneous rocks by alternating magnetic fields. Phil. Mag. **6**, pp. 225–241.

LAROCHELLE, A. 1966. Palaeomagnetism of the Abitibi dyke swarm. Can. J. Earth Sci. **3**, pp. 671–683.

LARSON, E., OZIMA, M., NAGATA, T., and STRANGWAY, D. 1969. Stability of remanent magnetization of igneous rocks. Geophys. J. Royal Astr. Soc. **17**, pp. 263–292.

LARSON, E. E. and STRANGWAY, D. W. 1969. Magnetization of the Spanish Peaks dike swarm, Colorado, and Shiprock dike, New Mexico. J. Geophys. Res. **74**, pp. 1505–1514.

MCELHINNY, M. W. 1963. Rock magnetism data reduction. Department of Geological Engineering, Princeton University, Princeton, New Jersey. Report 63–3.

——— 1966. An improved method for demagnetizing rocks in alternating magnetic fields. Geophys. J. Royal Astr. Soc. **10**, pp. 369–374.

MCELHINNY, M. W. and GOUGH, D. I. 1963. The palaeomagnetism of the Great dyke of Southern Rhodesia. Geophys. J. Royal Astr. Soc. **7**, pp. 287–303.

STRANGWAY, D. W. 1961. Magnetic properties of diabase dikes. J. Geophys. Res. **66**, pp. 3021–3032.

——— 1964. Rock magnetism and dike classification. J. Geol. **72**, pp. 648–663.

TARLING, D. H. and SYMONS, D. T. A. 1967. A stability index of remanence in palaeomagnetism. Geophys. J. Royal Astr. Soc. **12**, pp. 443–448.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and EDMONDS, C. M. 1967. Age determinations and geologic studies: K–Ar Isotopic ages, report 7. Geol. Surv. Can. Paper 66–17.

WATSON, G. S. 1956. Analysis of dispersion on a sphere. Mon. Not. Royal Astr. Soc. Geophys. Supp. **7**, pp. 153–159.

WILSON, M. E. 1965. The Precambrian of Canada; the Canadian Shield. *In* The Geologic Systems; the Precambrian. *Edited by* K. Rankama. Interscience, New York. 454 p.

WILSON, R. L., HAGGERTY, S. E., and WATKINS, N. D. 1968. Variation of palaeomagnetic stability and other parameters in a vertical traverse of a single Icelandic lava. Geophys. J. Royal Astr. Soc. **16**, pp. 79–96.